



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	MATERIAL SCIENCE				
Type of Course	ELECTIVE IN MAJOR				
Semester	VI				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Basics of solid-state chemistry 2. Brief idea about material properties				
Course Summary	The course in Material Science provides a comprehensive overview of various materials and their properties, focusing on applications in different fields. It covers the classification of materials based on structure and function, mechanical properties, testing methods, and specialized materials such as ferroelectric, piezoelectric, magnetic, and superconducting materials. Additionally, it discusses composite materials, including their definition, classification, processing methods, and applications. Through this course, students gain a solid understanding of different materials and their significance in modern technology.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO 1	Develop an understanding of the classification of materials based on structure and function.	U	F	Instructor-created exams / Quiz /Assignment
CO 2	Acquire knowledge of mechanical properties and testing methods to evaluate material behavior.	U	C	Class test /Assignment /Quiz
CO 3	Explore specialized materials such as ferroelectric, piezoelectric, magnetic, and	E	C	Assignment/ Class test

	superconducting materials and their applications.			
CO 4	Gain insight into composite materials, including their classification, processing techniques, and real-world applications.	E	P	Assignments /Seminar presentation
CO 5	Develop the ability to analyze and predict material behavior in various environments and applications	E	M	Assignments /Seminar presentation
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Introduction to Material Science		12	20
	1	Scope and importance of materials science. Classification of materials. Functional classification. Classification based on structure.	2	
	2	Mechanical properties – significance and terminology, the tensile test, true stress and true strain, bend test, hardness of materials.	3	
	3	Definition of ceramics. Traditional and new ceramics. Structure of ceramics. Atomic interactions and types of bonds.	2	
	4	Phase equilibria in ceramic systems, one component and multi component systems.	2	
	5	Use of phase diagrams in predicting material behavior.	1	
	6	Electrical, Magnetic, and Optical properties of ceramic materials.	2	
II	Materials for Special Purposes – I		12	20
	7	Production of ultra-pure materials - zone refining, vacuum distillation and electro refining.	2	
	8	Ferroelectric and piezoelectric materials - general properties. Classification of ferroelectric materials.	2	

	9	Theory of ferroelectricity, ferroelectric domains, applications.	3	
	10	Theory of Piezoelectricity. Piezoelectric materials and applications.	3	
	11	Metallic glasses - preparation, properties and applications.	2	
III	Materials for Special Purposes – II		12	20
	12	Magnetic materials, ferri, ferro and antiferromagnetism.	2	
	13	Metallic magnets, soft, hard & superconducting magnets.	2	
	14	Ceramic magnets, low conducting and superconducting magnets.	2	
	15	Superconducting materials - metallic and ceramic superconducting materials.	2	
	16	Theories of superconductivity, Meissner effect.	2	
	17	High temperature superconductors - structure and applications.	2	
IV	Composite Materials		12	20
	18	Definition and classification of composites, fibres and matrices.	2	
	19	Composites with metallic matrices – processing, solid and liquid state processing, deposition.	3	
	20	Ceramic matrix composite materials – processing, mixing & Pressing, liquid state processing, sol-gel processing & vapor deposition technique.	3	
	21	Interfaces in composites - mechanical & microstructural characteristics.	2	
	22	Applications of composites.	2	
V	Materials for Energy Harvesting and Storage		12	20
	Open Ended	Detailed study of materials used in data storage devices, light harvesting, energy storage, lasers and bioengineering.	12	

References:

1. W.D. Eingery, H.K. Downen and R.D. Uhlman, Introduction to Ceramics, John Wiley.
2. A.G. Guy, Essentials of Material Science, McGraw Hill.
3. M.J. Starfield and Shrager, Introductory Material Science, McGraw Hill.
4. S.K. Hajra Choudhary, Material Science and Engineering, Indian Book Dist. Co., Calcutta.
5. M.W. Barsoum, Fundamentals of Ceramics, McGraw Hill, 1997.
6. M. Tinkham, Introduction to Superconductivity, McGraw Hill, 1975.
7. A.V. Narlikar and A.N. Endnote, Superconductivity and Superconducting Materials, South

Asian Publishers, New Delhi, 1983.

8. S.V. Subramanyan and E.S. Rajagopal, High Temperature Superconductors, Wiley Eastern Ltd., 1988.

9. Azaroff and Brophy, Electronic Processes in Materials, McGraw Hill, 1985.

10. C.M. Srivastava and C. Srinivasan, Science of Engineering Materials, Wiley Eastern Ltd., 1987. R.J. Young, Introduction to Polymer Science, John Wiley and Sons.

11. K.K. Chowla, Composite Materials, Springer-Verlag, NY, 1987.

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	1					1	3				3		1
CO 2					1	2	3		1		2		1
CO 3				1	3	1	3		1		2		3
CO 4				2	3	1	3		1		2		3
CO 5				3		3	3		1		2		3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/Pr actical Exam	Assignmen t /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓